Group theory basics: orders, homomorphisms, isomorphism invariants

Notes from Carlos' lessons

Source lesson: 2026-01-28 (classification of groups).

Definitions (from the board)

A group $(G, \cdot)$ is associative, has an identity e ($ge = eg = g$), and every element has an inverse ($g^{-1}g = gg^{-1} = e$). It is not necessarily commutative ($ab \neq ba$ may happen).

- **Order of the group** $|G|$: the number of elements.
- **Order of an element** $\mathrm{ord}(g)$: the least $k \geq 1$ with $g^k = e$, or ∞ if no such k exists.
- Small examples on the board: $\mathbb{Z}_1, \ldots, \mathbb{Z}_4$ under $+$, with element orders (1), $(1, 2)$, $(1, 3, 3)$, $(1, 4, 2, 4)$; and $(\mathbb{R}, +)$, where every non-zero element has order ∞.

Cancellation (the "sudoku property")

Claim 1. *$gh = gk \Rightarrow h = k$, and likewise on the right. So every row and column of the multiplication table of a finite group contains each element exactly once.*

Proof. Multiply on the left by g^{-1}. □

Every element of a finite group has finite order

Theorem 2. *If G is finite, every $g \in G$ has* $\mathrm{ord}(g) \leq |G|$.

Proof. Consider $g^1, g^2, \ldots, g^{|G|+1}$: that is $|G| + 1$ elements of a set of size $|G|$, so by pigeonhole $g^i = g^j$ for some $1 \leq i < j \leq |G| + 1$. Cancelling g^i gives $g^{j-i} = e$ with $1 \leq j - i \leq |G|$. □

(The board restricts to $g^1, \ldots, g^{|G|}$ and excludes the identity to reach the same conclusion; the version above is the cleanest.)

Cyclic behaviour

For a fixed g the powers g^n either exhaust G or they do not.

- **Finite G, powers exhaust G:** $g^n \leftrightarrow n$ matches G with $(\mathbb{Z}_{|G|}, +)$.
- **Infinite G, powers exhaust G:** $g^n \leftrightarrow n$ matches G with $(\mathbb{Z}, +)$ (the board lists $\ldots, g^{-2}$, $g^{-1}, g^0, g^1, g^2, \ldots$).

The matching is a group isomorphism in the sense below, because $g^i g^j = g^{i+j}$.

Homomorphisms

A **homomorphism** $\varphi : (G, \cdot) \to (H, *)$ is a map with $\varphi(ab) = \varphi(a) * \varphi(b)$ (it "preserves structure"; "relabelling" on the board).

Claim 3 (Consequences)**.** *1.* $\varphi(e_G) = e_H$.
2. $\varphi(g^{-1}) = \varphi(g)^{-1}$.

Proof. (1) $\varphi(e_G) = \varphi(e_G e_G) = \varphi(e_G)\varphi(e_G)$; cancel $\varphi(e_G)$ in H. (2) $\varphi(g)\varphi(g^{-1}) = \varphi(gg^{-1}) = \varphi(e_G) = e_H$. □

An **isomorphism** is a homomorphism with an inverse homomorphism ($\varphi^{-1} \circ \varphi = \mathrm{id}_G$); equivalently, a bijective homomorphism.

Invariants: orders

Claim 4. *If $\varphi : G \to H$ is an isomorphism, then g and $\varphi(g)$ have the same order.*

Proof. $g^k = e_G \iff \varphi(g^k) = e_H \iff \varphi(g)^k = e_H$. The first "$\Leftarrow$" uses injectivity (only e_G maps to e_H); the second equivalence is the homomorphism property. □

Consequence (the lesson's strategy for classifying groups). For a group G, compute the multiset $S(G)$ of the orders of its elements. Isomorphic groups have the same $S(G)$, so different multisets prove that two groups are *not* isomorphic. For instance $S(\mathbb{Z}_3) = \{1, 3, 3\} \neq S(\mathbb{Z}_4) = \{1, 4, 2, 4\}$.

Notes

- The board is a definitions-and-first-lemmas lesson; the pigeonhole proof and the order invariant are the only real proofs, and both are given above.
- The board only claims that element orders are *an* invariant under isomorphism; it does not claim that $S(G)$ determines the group.